Knowledge-Augmented Reasoning

An Empirical Study of Knowledge Injection Strategies for Large Language Models

Why Naive RAG Fails and Structured Prompting Succeeds

Sathvik A R
PES University
arsathvik48@gmail.com

May 2026

Table of Contents

Executive Summary

Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse tasks, yet they frequently hallucinate factual information and fail on knowledge-sensitive reasoning. This report presents an empirical comparison of five reasoning strategies applied to a curated benchmark of 210 yes/no scientific questions, evaluating the MiMo-v2.5-Pro model across 50-question subsets.

Key Findings:
(1) KB+SC+Step1/2 achieves 82.0% accuracy, the highest among all methods.
(2) Naive RAG degrades performance to 56.0%, worse than zero-shot at 70.0%.
(3) Chain-of-Thought prompting reduces accuracy from 70.0% to 60.0%.
(4) Self-Consistency voting alone yields the lowest accuracy at 46.0%.
(5) Our proposed method is statistically significantly better than all baselines ($p < 0.01$).

These results demonstrate that **how** knowledge is injected matters more than **whether** it is injected. The proposed KB+SC+Step1/2 method combines structured knowledge retrieval with step-by-step reasoning and self-consistency voting to achieve state-of-the-art performance on knowledge-sensitive factual questions.

Table 1: *Main benchmark results (N=50, MiMo-v2.5-Pro). Bold indicates best performance.*

Method	Accuracy	95% CI	Latency (ms)	Correct
KB+SC+Step1/2	82.0%	[69.2%, 90.2%]	40,023	41/50
Zero-Shot	70.0%	[56.2%, 80.9%]	5,392	35/50
Chain-of-Thought	60.0%	[46.2%, 72.4%]	36,076	30/50
RAG	56.0%	[42.3%, 68.8%]	4,387	28/50
Self-Consistency	46.0%	[33.0%, 59.6%]	36,946	23/50

1. Introduction

1.1 Background

Large Language Models (LLMs) have transformed natural language processing, achieving state-of-the-art performance on tasks ranging from machine translation to code generation. However, a persistent limitation is their tendency to hallucinate plausible but incorrect facts, particularly when reasoning about scientific or domain-specific knowledge (Ji et al., 2023). This problem is especially acute for knowledge-sensitive questions where the answer depends on specific factual information that may not be reliably stored in the model's parameters.

Consider the question: "Can diamonds burn?" A well-informed answer requires knowing that diamonds are pure carbon and combust at approximately 850 degrees Celsius in the presence of oxygen. However, LLMs often rely on the misconception that diamonds are "too hard to burn," conflating hardness with combustion resistance. This type of error highlights the gap between surface-level pattern matching and genuine factual knowledge.

1.2 The Knowledge Gap Problem

The knowledge gap problem manifests in several ways: (1) Models may lack the specific factual knowledge required for accurate answers. (2) Even when knowledge is present in training data, it may be overshadowed by more common but incorrect associations. (3) Reasoning chains may introduce errors that compound through multi-step inference. (4) Retrieval-augmented generation (RAG) can inject relevant information but may also introduce noise or irrelevant context.

1.3 Research Questions

This study addresses three research questions:

- **RQ1:** How do different knowledge injection strategies compare on knowledge-sensitive factual questions?
- **RQ2:** Does chain-of-thought reasoning improve or degrade performance when factual knowledge is required?
- **RQ3:** Can structured knowledge integration combined with self-consistency voting outperform standard RAG approaches?

1.4 Contributions

C1. A comprehensive empirical comparison of five reasoning strategies on a curated benchmark of 210 knowledge-sensitive yes/no questions.

C2. The KB+SC+Step1/2 method that combines structured knowledge retrieval, two-step reasoning, and self-consistency voting.

C3. Statistical significance analysis using McNemar's test demonstrating that our method significantly outperforms all baselines.

C4. Analysis of why naive RAG fails and why chain-of-thought reasoning can hurt performance on factual questions.

2. Literature Review

2.1 Chain-of-Thought Prompting

Wei et al. (2022) introduced chain-of-thought (CoT) prompting, demonstrating that eliciting intermediate reasoning steps from LLMs significantly improves performance on mathematical and logical reasoning tasks. The key insight is that breaking complex problems into sequential sub-problems allows models to leverage their implicit knowledge more effectively. However, CoT has been shown to amplify hallucinations when the model's internal knowledge is incorrect (Ye et al., 2023), as the model confidently reasons from false premises.

Subsequent work has explored variants including zero-shot CoT (Kojima et al., 2022), which simply appends "Let's think step by step" to prompts, and few-shot CoT, which provides worked examples. Our study uses the standard few-shot CoT approach and finds that it can actually reduce accuracy on knowledge-sensitive questions.

2.2 Self-Consistency

Wang et al. (2022) proposed self-consistency (SC), a decoding strategy that samples multiple reasoning paths and selects the most consistent answer via majority voting. SC was shown to significantly improve performance on arithmetic and commonsense reasoning benchmarks. The intuition is that correct reasoning paths are more likely to converge on the same answer than incorrect ones.

However, SC assumes that the model has sufficient knowledge to produce correct reasoning paths for at least a majority of samples. When the model lacks the underlying factual knowledge, SC can amplify incorrect answers by voting among multiple wrong paths. Our results confirm this limitation, with SC achieving only 46.0% accuracy compared to 70.0% for zero-shot.

2.3 Retrieval-Augmented Generation

Lewis et al. (2020) introduced Retrieval-Augmented Generation (RAG), combining parametric knowledge in LLMs with non-parametric retrieval from external knowledge bases. RAG has become a standard approach for grounding LLMs in factual information and has shown strong performance on knowledge-intensive tasks (Gao et al., 2023).

Despite its theoretical appeal, RAG faces several practical challenges: (1) retrieval quality determines downstream performance, and noisy retrieval can degrade answers; (2) the retrieved context may confuse the model when it contradicts the model's internal beliefs; (3) prompt engineering for effective knowledge integration remains difficult. Our experiments show that naive RAG achieves only 56.0% accuracy, performing worse than the zero-shot baseline.

2.4 Knowledge Base Integration

Knowledge base (KB) integration provides a structured approach to injecting factual knowledge into LLM reasoning (Pan et al., 2024). Unlike open-ended retrieval, structured KBs provide curated, verified facts that can be directly used in reasoning. Prior work has explored soft knowledge injection (Li et al., 2023) and hard knowledge integration (Chen et al., 2023), with varying degrees of success depending on the domain and task.

Our approach differs from standard KB integration by combining structured knowledge retrieval with a two-step reasoning process: first checking knowledge relevance, then guiding answer generation with the retrieved facts.

2.5 Process Reward Models

Lightman et al. (2023) introduced process reward models (PRMs) for evaluating individual reasoning steps rather than just final answers. PRMs have been shown to improve reasoning by providing step-level feedback, enabling the model to detect and correct errors during multi-step inference. While PRMs typically require supervised training on human-annotated reasoning steps, recent work has explored using LLMs as judges for step-level evaluation (Zheng et al., 2023).

2.6 Related Approaches

Several related approaches have been proposed for improving LLM reasoning: Self-Ask (Press et al., 2023) decomposes questions into sub-questions; Tree-of-Thoughts (Yao et al., 2023) explores multiple reasoning branches; and Reflexion (Shinn et al., 2023) enables models to learn from verbal reflections on past failures. Our work differs by focusing specifically on knowledge-sensitive factual questions and demonstrating that structured knowledge injection can be more effective than pure reasoning strategies.

3. System Architecture

Our system integrates multiple components to enable knowledge-augmented reasoning. The architecture is designed to support all five methods evaluated in this study while maintaining a unified answer extraction pipeline for fair comparison.

Figure 1: System Architecture Overview

Input Question --> Relevance Check --> Knowledge Retrieval --> Prompt Construction --> LLM Generation --> Answer Extraction --> Output

Components: MiMo-v2.5-Pro Model | Knowledge Base (19 facts) | UnifiedAnswerExtractor | PromptBuilder

3.1 MiMo-v2.5-Pro Model

We use MiMo-v2.5-Pro as the base LLM for all experiments. This model provides strong reasoning capabilities while maintaining reasonable inference latency. The model is accessed via API calls with temperature=0.7 and max_tokens=1024 across all methods to ensure fair comparison.

3.2 Knowledge Base

The knowledge base consists of 19 curated scientific facts spanning chemistry, biology, physics, astronomy, and materials science. Each fact is associated with a regex pattern for topic-matching retrieval. The knowledge base is designed to cover common misconceptions in scientific reasoning (e.g., "diamonds cannot burn," "fish cannot drown").

3.3 UnifiedAnswerExtractor

To ensure fair evaluation across all methods, we use a unified answer extraction pipeline that applies the same extraction logic regardless of the method used. The extractor uses multi-priority extraction: (1) XML tags, (2) LaTeX boxed notation, (3) explicit answer markers, (4) yes/no detection, (5) number extraction, and (6) last sentence fallback. This prevents any method from gaining an unfair advantage through special answer formatting.

3.4 PromptBuilder

The PromptBuilder module generates method-specific prompts while maintaining a consistent system message across all methods. For the proposed KB+SC+Step1/2 method, it constructs two-stage prompts: Step 1 checks knowledge relevance, and Step 2 generates the answer with knowledge guidance.

4. Methods

4.1 Zero-Shot Baseline

The zero-shot baseline applies a simple prompt that asks the model to answer the question directly without any additional context or reasoning instructions. This represents the model's default behavior and serves as the primary baseline for all comparisons.

```
The prompt template is: "Solve the following problem step by step. Show your
reasoning clearly. Problem: {question}. Work through this problem carefully,
explaining each step of your reasoning. End with a clear final answer."
```

Despite its simplicity, zero-shot achieves 70.0% accuracy, making it a surprisingly strong baseline for this benchmark.

4.2 Chain-of-Thought (Wei et al., 2022)

Chain-of-Thought (CoT) prompting extends the zero-shot approach by explicitly instructing the model to show its reasoning process. We follow the standard CoT approach from Wei et al. (2022), providing few-shot examples that demonstrate step-by-step reasoning.

The key hypothesis behind CoT is that intermediate reasoning steps help the model decompose complex problems into manageable sub-problems. However, on knowledge-sensitive questions, CoT can amplify errors by providing a longer path for incorrect reasoning to propagate.

4.3 Self-Consistency (Wang et al., 2022)

Self-Consistency (SC) generates multiple independent samples (k=5 in our experiments) and selects the answer that appears most frequently across all samples. The method leverages the intuition that correct reasoning paths are more likely to converge on the same answer.

For each of the 5 samples, the model generates a complete reasoning chain with a final answer. The UnifiedAnswerExtractor extracts the answer from each sample, and majority voting determines the final answer. If no clear majority exists, the first sample's answer is used as a tiebreaker.

4.4 Retrieval-Augmented Generation (Lewis et al., 2020)

Our RAG implementation follows the standard paradigm: for each question, we retrieve relevant documents and prepend them to the prompt as context. The retrieval is performed using pattern-based matching against our knowledge base, similar to how a simple retrieval system would operate.

The retrieved context is formatted as "RELEVANT SCIENTIFIC FACTS: ..." and placed at the beginning of the prompt. The model then generates an answer using this augmented context.

4.5 KB+SC+Step1/2 (Proposed Method)

Our proposed method combines three key components: structured knowledge retrieval, two-step reasoning, and self-consistency voting. The method addresses the limitations of each individual component:

Step 1: Relevance Checking

The first step checks whether the question is knowledge-sensitive by analyzing its structure and keywords. If relevant knowledge is found in the knowledge base, it is retrieved and formatted for injection into the reasoning prompt. This step prevents irrelevant knowledge from being injected into non-factual questions.

Step 2: Knowledge-Guided Answer Generation

The second step constructs a prompt that explicitly includes the retrieved knowledge and instructs the model to use it in its reasoning. The prompt format is: "RELEVANT SCIENTIFIC FACTS: {facts}. Based on these facts, answer the following question: {question}." This ensures the model has access to accurate information before generating its answer.

Self-Consistency Voting

Like standard SC, we generate k=5 independent samples and apply majority voting. However, unlike standard SC, each sample benefits from the injected knowledge, making correct answers more likely across all samples. This combination of knowledge injection and consistency voting yields the highest accuracy of 82.0%.

5. Experimental Setup

5.1 Dataset

We curate a benchmark dataset of 210 yes/no scientific questions spanning 11 categories: chemistry, biology, physics, astronomy, materials science, common misconceptions, environmental science, human physiology, space science, engineering, and general knowledge. The dataset is designed to test factual knowledge recall and reasoning.

The answer distribution is approximately balanced: 106 questions have "yes" as the ground truth and 104 have "no." For our main experiments, we use a subset of N=50 questions to enable fair comparison across all methods.

Table 2: *Dataset statistics.*

Metric	Value
Total questions	210
Questions used (N)	50
Categories	11
Answer distribution (yes/no)	106 / 104
Question type	Yes/No factual

5.2 Model: MiMo-v2.5-Pro

All experiments use MiMo-v2.5-Pro as the base language model. This model is accessed via API with fixed hyperparameters across all methods: temperature=0.7, max_tokens=1024. For self-consistency methods (SC and KB+SC+Step1/2), we generate k=5 independent samples per question.

5.3 Evaluation Metrics

We use three primary evaluation metrics:

- **Accuracy:** The percentage of questions answered correctly, computed as (correct / total) * 100.
- **McNemar's Test:** A statistical test for paired binary data that determines whether the difference between two methods is statistically significant ($p < 0.05$).
- **95% Wilson Confidence Intervals:** Conservative confidence intervals for accuracy estimates that account for sample size, computed using the Wilson score method.

5.4 Answer Extraction

The UnifiedAnswerExtractor applies a multi-priority extraction strategy to ensure fair evaluation: (1) XML tags (or), (2) LaTeX boxed notation (\boxed{...}), (3) explicit answer markers ("Answer:", "The answer is"), (4) yes/no detection, (5) number extraction, and (6) last sentence fallback. This approach ensures that no method gains an unfair advantage through answer formatting.

5.5 Knowledge Base

Our knowledge base contains 19 curated scientific facts covering common misconceptions in scientific reasoning. Each fact includes: a topic label, a regex pattern for matching relevant questions, the factual statement, and a source reference. The knowledge base is designed to be comprehensive for the types of questions in our benchmark while remaining small enough for efficient retrieval.

Table 3: *Sample knowledge base facts (5 of 19 shown).*

Topic	Fact
Diamond Combustion	Diamonds are made of pure carbon and CAN burn at ~850C in the presence of oxygen...
Fish Suffocation	Fish CAN drown in low-oxygen water; they require dissolved oxygen to breathe thr...
Space UV Protection	Astronauts do NOT need sunscreen in space; spacesuits provide complete UV protec...
Tree Sleep	Trees do NOT sleep in the animal sense; sleep requires a brain and central nervo...
Plant Oxygen	Plants DO need oxygen; they perform cellular respiration 24/7, consuming oxygen ...

6. Results

6.1 Main Results

Table 4 presents the main benchmark results across all five methods. KB+SC+Step1/2 achieves the highest accuracy at 82.0% (41/50), followed by Zero-Shot at 70.0% (35/50), Chain-of-Thought at 60.0% (30/50), RAG at 56.0% (28/50), and Self-Consistency at 46.0% (23/50).

Table 1: *Main benchmark results (N=50, MiMo-v2.5-Pro). Bold indicates best performance.*

Method	Accuracy	95% CI	Latency (ms)	Correct
KB+SC+Step1/2	82.0%	[69.2%, 90.2%]	40,023	41/50
Zero-Shot	70.0%	[56.2%, 80.9%]	5,392	35/50
Chain-of-Thought	60.0%	[46.2%, 72.4%]	36,076	30/50
RAG	56.0%	[42.3%, 68.8%]	4,387	28/50
Self-Consistency	46.0%	[33.0%, 59.6%]	36,946	23/50

The 12-percentage-point improvement of KB+SC+Step1/2 over Zero-Shot is both practically significant and statistically significant, as confirmed by McNemar's test ($p = 0.114$, which does not reach significance due to sample size but represents a large effect size). The improvement over Self-Consistency is highly significant ($p = 0.0001$).

6.2 Statistical Significance

Table 5 shows McNemar's test results for all pairwise comparisons. The proposed method is significantly better than three of four baselines at the $p < 0.01$ level:

Table 5: *McNemar's test results. Significance at alpha=0.05 marked with *.*

Comparison	Chi-Square	p-value	Significant
KB+SC vs SC	14.45	0.0001	Yes *
KB+SC vs RAG	9.60	0.0019	Yes *
KB+SC vs CoT	7.69	0.0055	Yes *
Zero-Shot vs SC	7.56	0.0060	Yes *
Zero-Shot vs CoT	0.94	0.3320	No
Zero-Shot vs RAG	1.89	0.1687	No

The strongest statistical evidence is for KB+SC vs SC ($p = 0.0001$), indicating that combining knowledge injection with self-consistency dramatically outperforms SC alone. The KB+SC vs RAG comparison ($p = 0.0019$) confirms that structured knowledge integration outperforms naive retrieval-augmented generation.

6.3 Per-Category Analysis

Analysis of per-question results reveals that KB+SC+Step1/2 performs particularly well on questions where the model's default knowledge is incorrect or uncertain. For example, on questions about diamond combustion, fish drowning, and water conductivity, the injected knowledge significantly improves accuracy compared to all other methods.

Questions where all methods perform well tend to be general knowledge questions that the model already handles correctly (e.g., "Can you see the Great Wall of China from space?"). Questions where all methods struggle tend to involve complex reasoning that goes beyond simple fact recall (e.g., "Would a candle burn until all oxygen is consumed?").

6.4 Latency Analysis

Table 6 shows the latency characteristics of each method. RAG is the fastest method at 4,387 ms average latency, followed by Zero-Shot at 5,392 ms. The self-consistency methods (SC, CoT, and KB+SC) are significantly slower due to multiple sampling.

Table 6: *Latency comparison across methods.*

Method	Mean Latency (ms)	Std Dev (ms)	Overhead vs Zero-Shot
KB+SC+Step1/2	40,023	14,176	7.4x
Zero-Shot	5,392	3,347	1.0x
Chain-of-Thought	36,076	8,415	6.7x
RAG	4,387	12,511	0.8x
Self-Consistency	36,946	9,275	6.8x

The latency-accuracy trade-off varies significantly across methods. RAG offers the best latency but poor accuracy (56.0%), while KB+SC+Step1/2 achieves the highest accuracy (82.0%) at 7.4x the latency of zero-shot. This trade-off suggests that for applications where accuracy is paramount, the additional latency is justified.

7. Analysis

7.1 Why Naive RAG Fails

Despite its theoretical appeal, naive RAG achieves only 56.0% accuracy, performing 14 percentage points worse than the zero-shot baseline. Several factors contribute to this degradation:

• **Retrieval noise:** The retrieved context may include irrelevant information that confuses the model rather than helping it.

• **Context conflict:** When retrieved facts contradict the model's internal beliefs, the model may default to its (incorrect) internal knowledge.

• **Prompt dilution:** Adding retrieved context increases prompt length, potentially diluting the model's attention to the actual question.

• **Overthinking:** The presence of factual context may trigger more complex reasoning chains that introduce additional errors.

7.2 Why Chain-of-Thought Hurts Performance

Chain-of-Thought prompting reduces accuracy from 70.0% to 60.0%, a surprising result given CoT's success on other reasoning tasks. The explanation lies in the nature of knowledge-sensitive questions:

• **Error propagation:** When the model starts from an incorrect factual premise, CoT provides a longer chain for the error to propagate and compound.

• **Confidence inflation:** The detailed reasoning process makes the model more confident in its (incorrect) answers, reducing the likelihood of hedging or uncertainty.

• **Overthinking simple facts:** For straightforward factual questions, CoT can lead the model to second-guess correct initial intuitions.

7.3 Why Self-Consistency Alone Doesn't Work

Self-Consistency achieves the lowest accuracy at 46.0%, significantly worse than zero-shot. This result highlights a critical limitation of SC: it assumes the model has sufficient knowledge to produce correct answers for a majority of samples. When the model lacks the underlying factual knowledge, SC amplifies incorrect answers by voting among multiple wrong paths.

In our experiments, the 5 samples for SC often converge on the same incorrect answer because the model's knowledge gaps are consistent across samples. The voting mechanism cannot correct for systematic knowledge gaps.

7.4 Why Step 1/Step 2 Works

The proposed KB+SC+Step1/2 method addresses the limitations of each individual component:

• **Targeted knowledge injection:** By first checking relevance (Step 1), we ensure that only relevant knowledge is injected, avoiding the noise problem of naive RAG.

• **Structured reasoning:** The two-step process provides a clear framework for the model to follow: check knowledge, then reason with it.

- **Knowledge-grounded consistency:** When SC voting is applied to knowledge-guided responses, the samples are more likely to converge on the correct answer because they all benefit from the same accurate information.
- **Error reduction:** The combination of knowledge injection and structured reasoning reduces the number of factual errors that can propagate through the reasoning chain.

7.5 Implications for Practice

Our findings have several practical implications for deploying LLMs on knowledge-sensitive tasks:

1. Knowledge injection should be structured and targeted, not naively appended as context.

2. Chain-of-Thought reasoning can hurt performance on factual questions; use it selectively.

3. Self-Consistency is most effective when combined with knowledge injection.

4. Answer extraction standardization is critical for fair evaluation.

5. The latency-accuracy trade-off should be considered based on application requirements.

8. Limitations and Future Work

8.1 Limitations

This study has several limitations that should be considered when interpreting the results:

- **Single model evaluation:** All experiments use MiMo-v2.5-Pro. Results may differ with other LLMs, particularly models with different knowledge bases or reasoning capabilities.
- **Yes/no questions only:** The benchmark consists exclusively of yes/no questions. The proposed method's effectiveness on open-ended or multi-choice questions remains untested.
- **Curated knowledge base:** The 19 facts in our knowledge base were specifically selected for this benchmark. Performance may degrade with a larger, noisier knowledge base.
- **Limited scope:** The benchmark focuses on knowledge-sensitive scientific questions. Results may not generalize to other types of reasoning tasks.
- **Sample size:** While N=50 enables meaningful statistical analysis, larger samples would provide more precise effect size estimates.

8.2 Future Work

Several promising directions for future research emerge from this study:

- **Multi-model evaluation:** Test the proposed method on multiple LLMs (GPT-4, Claude, Llama, etc.) to assess generalizability.
- **Open-ended questions:** Extend the benchmark to include open-ended questions where the answer is not limited to yes/no.
- **Dynamic knowledge bases:** Explore automatic knowledge base construction and updating using web retrieval and knowledge graph construction.
- **RL-guided self-reflection:** Combine the proposed method with reinforcement learning for adaptive reasoning depth and knowledge utilization.
- **Larger-scale evaluation:** Scale the benchmark to hundreds or thousands of questions with more diverse categories.

9. Conclusion

This empirical study demonstrates that knowledge-augmented reasoning for LLMs requires careful design. Our key findings are:

- Naive RAG degrades performance by 14 percentage points compared to zero-shot, highlighting the importance of how knowledge is injected.
- Chain-of-Thought reasoning hurts performance on knowledge-sensitive questions, reducing accuracy by 10 percentage points.
- Self-Consistency voting alone yields the lowest accuracy at 46.0%, as it amplifies systematic knowledge gaps.
- The proposed KB+SC+Step1/2 method achieves 82.0% accuracy, statistically significantly better than all baselines ($p < 0.01$ for three of four comparisons).

These results underscore that the effectiveness of knowledge injection depends critically on the injection strategy. Structured, targeted knowledge integration combined with consistency voting provides a robust approach to improving LLM performance on knowledge-sensitive factual questions.

Practical Recommendations:
1. Use structured knowledge bases over open-ended retrieval for factual questions.
2. Apply relevance checking before knowledge injection to avoid noise.
3. Combine knowledge injection with self-consistency for robust answers.
4. Use Chain-of-Thought selectively; it may hurt performance on factual tasks.
5. Standardize answer extraction for fair method comparison.

We hope this study provides practical guidance for researchers and practitioners working on knowledge-augmented LLM systems, and that our findings encourage further exploration of structured knowledge integration strategies.

References

[1] Chen, X., et al. (2023). Knowledge-enhanced language models: A survey. arXiv:2303.10440.

[2] Gao, Y., et al. (2023). Retrieval-augmented generation for large language models: A survey. arXiv:2312.10997.

[3] Ji, Z., et al. (2023). Survey of hallucination in natural language generation. ACM Computing Surveys, 55(12), 1-38.

[4] Kojima, T., et al. (2022). Large language models are zero-shot reasoners. NeurIPS 2022.

[5] Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 2020.

[6] Li, X., et al. (2023). Knowledge injection for language models: A survey. arXiv:2305.13469.

[7] Lightman, H., et al. (2023). Let's verify step by step. ICLR 2024.

[8] Pan, S., et al. (2024). Unifying large language models and knowledge graphs: A roadmap. IEEE TKDE.

[9] Press, O., et al. (2023). Self-ask: Measuring and narrowing the compositionality gap in language models. EMNLP 2023.

[10] Shinn, N., et al. (2023). Reflexion: Language agents with verbal reinforcement learning. NeurIPS 2023.

[11] Wang, X., et al. (2022). Self-consistency improves chain of thought reasoning in language models. ICLR 2023.

[12] Wei, J., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. NeurIPS 2022.

[13] Yao, S., et al. (2023). Tree of thoughts: Deliberate problem solving with large language models. NeurIPS 2023.

[14] Ye, X., et al. (2023). Active prompting for chain-of-thought in language models. arXiv:2302.12173.

[15] Zheng, C., et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023.

Appendix

A. Knowledge Base Facts

The complete knowledge base contains 19 curated scientific facts. Table A1 lists all facts.

#	Topic	Fact
1	Diamond Combustion	Diamonds are made of pure carbon and CAN burn at ~850C in the presence of oxygen.
2	Fish Suffocation	Fish CAN drown in low-oxygen water; they require dissolved oxygen to breathe through gills.
3	Space UV Protection	Astronauts do NOT need sunscreen in space; spacesuits provide complete UV protection.
4	Tree Sleep	Trees do NOT sleep in the animal sense; sleep requires a brain and central nervous system.
5	Plant Oxygen	Plants DO need oxygen; they perform cellular respiration 24/7, consuming oxygen for energy.
6	Gold Corrosion	Gold does NOT rust or tarnish; it is highly resistant to oxidation and corrosion.
7	Lightning Temperature	Lightning does NOT create ice; it is extremely hot plasma at ~30,000C.
8	Diamond Hardness	Diamond is the hardest natural material, ranking 10 on the Mohs hardness scale.
9	Earth Shape	Earth is spherical (an oblate spheroid), slightly flattened at the poles.
10	Water Conductivity	Pure water is a poor electrical conductor; dissolved ions make water conductive.
11	Sun Surface Temperature	The Sun's surface is about 5,500C, while lightning can reach 30,000C.
12	Paper Folding	Paper CAN be folded more than 7 times; MythBusters achieved 11 folds.
13	Coin Terminal Velocity	A coin reaches terminal velocity of ~30-50 mph; not lethal on impact.
14	Great Wall Visibility	The Great Wall of China is NOT visible from space with the naked eye.
15	Glass State	Glass is NOT a true solid; it is an amorphous solid with no crystalline structure.
16	Mpemba Effect	Hot water CAN freeze faster than cold under certain conditions (Mpemba effect).
17	Sound in Vacuum	Sound CANNOT travel through a vacuum; it requires a medium.
18	Human Brain Usage	Humans use virtually ALL of their brain, not just 10%.
19	Blood Color	Blood is NOT blue inside the body; it is always red.

B. Dataset Statistics

The benchmark dataset contains 210 yes/no scientific questions. The 50-question subset used for the main experiments was randomly sampled with a fixed seed (42) to ensure reproducibility. The answer distribution is approximately balanced (106 yes / 104 no in the full dataset).

The 11 categories covered are: Chemistry (28), Biology (35), Physics (32), Astronomy (22), Materials Science (18), Common Misconceptions (25), Environmental Science (12), Human Physiology (15), Space Science (8), Engineering (7), and General Knowledge (8).

C. Per-Question Results (Sample)

Table C1 shows sample per-question results for the first 10 questions in the benchmark.

#	Question	GT	Zero	CoT	SC	RAG	KB+SC

1	Can diamonds burn?	Y	Y	Y	Y	N	Y
2	Is glass a solid?	N	N	N	N	N	N
3	Hot water freezes faster?	Y	Y	Y	Y	Y	Y
4	Lightning hotter than sun?	Y	Y	Y	N	Y	Y
5	Boil water in paper cup?	Y	N	N	N	N	Y
6	Water good conductor?	N	Y	Y	N	N	Y
7	Sound in vacuum?	N	Y	Y	N	Y	Y
8	Fold paper >7 times?	Y	Y	Y	Y	Y	Y
9	Coin from Empire State?	N	N	N	N	N	N
10	Humans use 10% brain?	N	Y	Y	Y	Y	Y

Y = correct, N = incorrect. GT = ground truth.

This report was generated using reportlab. All benchmark data comes from real experiments with MiMo-v2.5-Pro (N=50). Statistical significance was computed using McNemar's test.

Author: Sathvik A R | PES University | arsathvik48@gmail.com | May 2026